Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_01

1. Summary of Key Deviations

The test run shows a complex pattern of disturbances affecting multiple subsystems:

- **Flow_PRV (Critical):** Massive and persistent **signal saturation** (+1000% to +11000%) across all states (s1–s7). This is the most dominant anomaly.
- **Flow_Bypass (Moderate):** Positive **offset and drift** (up to +11.4%) starting in s2 and persisting through s7.
- **p_Combustion (Moderate):** Positive **offset and drift** (up to +8.4%) from s2 onwards.
- **Flow_Engine1-4 (Minor):** Consistent **negative offset** (approx. -3.0% to -4.9%) during states s4–s6 (high flow/throttle phases).
- **Flow_Pump (Minor):** Slight positive **offset** (approx. +6%) in most states.
- **p_FMU / p_Shut (Minor):** Slight positive **drift/offset** in early states.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Flow_PRV Sensor**	**Sensor Saturation / Failure**	The deviation is extreme (+11000%) and physically impossible for a real flow increase in this system. It persists from s1 (idle) through s7, suggesting a signal-level failure rather than a physical flow event.
2	**Fuel Metering Unit (FMU)**	**Partial Blockage / Restricted Opening**	**Flow_Engine1-4** are all consistently lower than healthy (-3% to -5%) during throttle phases (s4-s6), while **p_Combustion** and **Flow_Bypass** are higher. This indicates flow is being restricted from entering the engines and is being diverted back through the bypass.

3	**Pressure Relief Valve (PRV)**	**Mechanical Jam (Partial Open)**	If the PRV is stuck slightly open, it would physically divert flow. However, the saturation signal is so high that it masks actual physical behavior. If the PRV were physically wide open, system pressures would likely drop, but here **p_Combustion** is high.
4	**Bypass Valve**	**Calibration Shift / Degradation**	The **Flow_Bypass** shows a steady positive offset. Combined with high **p_Combustion**, it suggests the bypass is opening more than intended or responding to higher system back-pressure.
5	**Gear Pump**	**Over-delivery / Efficiency Deviation**	**Flow_Pump** is +6% higher than healthy. This could be a secondary effect of high system pressure or a slight deviation in the pump's performance model, though it is less likely to be the root cause of the engine flow drop.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault: A Primary Sensor Failure of Flow_PRV and a Secondary Restriction in the FMU.**

Reasoning: The extreme saturation of **Flow_PRV** (+11248%) is a clear sensor-level failure; no physical component could produce such a massive positive deviation without the system self-destructing. However, this sensor fault alone does not explain why the **Flow_Engine** signals are low while **p_Combustion** and **Flow_Bypass** are high. This secondary pattern strongly suggests a physical restriction downstream of the pump—likely in the **FMU**—which limits flow to the nozzles and increases the pressure and bypass flow upstream.